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Introduction

This report presents the design and simulation of a multistage NMOS voltage amplifier implemented in KiCad. The objective of the design was to achieve a high voltage gain while maintaining proper biasing, acceptable bandwidth, and sufficient output swing.

To meet these requirements, a four-stage amplifier was constructed using three cascaded Common-Source (CS) stages followed by a final output stage. The amplifier operates from a 3.3 V supply and uses AC coupling between stages to isolate DC biasing. Simulation results show that the amplifier achieves a midband gain of approximately 61 dB and an output swing close to the required specification.

Architecture Justification

The amplifier is composed of three cascaded common-source (CS) stages followed by a common-drain (CD) stage. This configuration is selected to achieve high voltage gain through the CS stages while using the CD stage to provide low output impedance and effectively drive the load.

The first three stages (M1, M2, and M3) operate as common-source amplifiers. Each stage employs a resistive voltage divider at the gate to establish a stable DC bias point, ensuring operation in the saturation region. The drain resistors (RD1, RD2, and RD3) convert variations in drain current into output voltage, thereby producing voltage gain. AC coupling capacitors (Cin1, C1, and C2) are used between stages to block DC components while allowing the AC signal to propagate through the amplifier.

Since each common-source stage provides voltage gain, cascading multiple stages results in a significant increase in overall amplification. However, inter-stage loading reduces the effective gain compared to the ideal case, as each stage presents a finite input resistance to the previous stage.

The final stage (M4) is configured as a common-drain (source follower). This stage provides a voltage gain close to unity while significantly lowering the output impedance, enabling the amplifier to drive the load resistor (RL) with minimal signal attenuation. The output is taken from the source of M4, and the load capacitor (CL) models the effect of capacitive loading on the amplifier.

Overall, the multi-stage architecture achieves a balance between high voltage gain and effective load driving capability, with the CS stages providing amplification and the CD stage ensuring efficient signal delivery to the output.

Stage	Transistor	Topology	Gain (V/V)	Gain dB
1	M1	CS	12.6	22
2	M2	CS	12.5	22
3	M3	CS	12.6	22
4	M4	CD	0.6	-4.4

The stage gains were extracted from AC simulation at midband frequency. The first three common-source stages provide the majority of the gain, while the final common-drain stage has a gain less than unity and acts as a buffer.

Calculations

<u>Step 1 - M1 (Common - Source, $R_D = 25\text{ k}\Omega$)</u>	
<u>DC</u> $V_{GS} = V_{DD} \left(\frac{R_{A1b}}{R_{A1a} + R_{A1b}} \right) = 3.3 \frac{1\text{ M}}{2.7\text{ M} + 1\text{ M}} = 0.892\text{ V}$ $V_S = 0\text{ V}$ $V_{GS} = V_G - V_S = 0.892 - 0 = 0.892\text{ V}$ $V_{OV} = V_{GS} - V_{th} = 0.892 - 0.63 = 0.262\text{ V}$ $I_D = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right) V_{OV}^2 = \frac{1}{2} (100\mu\text{A}) \left(\frac{5}{1} \right) (0.262)^2 \approx 27.4\mu\text{A}$ $V_D = V_{DD} - I_D R_D = 3.3 - (27.4\mu\text{A}) (25\text{ k}) \approx 2.615\text{ V}$ $V_{DS} = V_D - V_S = 2.615 - 0 = 2.615\text{ V}$ Since: $V_{DS} > V_{OV}$ M_1 is in SAT.	<u>AC</u> $g_m = \frac{2I_D}{V_{OV}} = \frac{2(27.4\mu\text{A})}{0.262} \approx 0.209\text{ mA/V}$ $r_o = \frac{1}{\lambda I_D} = \frac{1}{(0.04)(27.4\mu\text{A})} \approx 912\text{ k}\Omega$ $R_D \parallel r_o = 25\text{ k} \parallel 912\text{ k} \approx 24.3\text{ k}\Omega$ $ A_{v1} = g_m (R_D \parallel r_o) = (0.209\text{ mA/V}) (24.3\text{ k}) \approx 5.08\text{ V/V}$ $20 \log(5.08) \approx 14.1\text{ dB}$
<u>Step 2 - M2 (Common - Source, $R_D = 25\text{ k}\Omega$)</u>	
<u>DC</u> $V_{GS} = V_{DD} \left(\frac{R_{A2b}}{R_{A2a} + R_{A2b}} \right) = 3.3 \frac{1\text{ M}}{2.7\text{ M} + 1\text{ M}} = 0.892\text{ V}$ $V_S = 0\text{ V}$ $V_{GS} = V_G - V_S = 0.892 - 0 = 0.892\text{ V}$ $V_{OV} = V_{GS} - V_{th} = 0.892 - 0.63 = 0.262\text{ V}$ $I_D = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right) V_{OV}^2 = \frac{1}{2} (100\mu\text{A}) \left(\frac{5}{1} \right) (0.262)^2 \approx 27.4\mu\text{A}$ $V_D = V_{DD} - I_D R_D = 3.3 - (27.4\mu\text{A}) (25\text{ k}) \approx 2.615\text{ V}$ $V_{DS} = V_D - V_S = 2.615 - 0 = 2.615\text{ V}$ Since: $V_{DS} > V_{OV}$ M_2 is in SAT.	<u>AC</u> $g_m = \frac{2I_D}{V_{OV}} = \frac{2(27.4\mu\text{A})}{0.262} \approx 0.209\text{ mA/V}$ $r_o = \frac{1}{\lambda I_D} = \frac{1}{(0.04)(27.4\mu\text{A})} \approx 912\text{ k}\Omega$ $R_D \parallel r_o = 25\text{ k} \parallel 912\text{ k} \approx 24.3\text{ k}\Omega$ $ A_{v2} = g_m (R_D \parallel r_o) = (0.209\text{ mA/V}) (24.3\text{ k}) \approx 5.08\text{ V/V}$ $20 \log(5.08) \approx 14.1\text{ dB}$
<u>Step 3 - M3 (Common - Source, $R_D = 25\text{ k}\Omega$)</u>	
<u>DC</u> $V_{GS} = V_{DD} \left(\frac{R_{A3b}}{R_{A3a} + R_{A3b}} \right) = 3.3 \frac{1\text{ M}}{2.7\text{ M} + 1\text{ M}} = 0.892\text{ V}$ $V_S = 0\text{ V}$ $V_{GS} = V_G - V_S = 0.892 - 0 = 0.892\text{ V}$ $V_{OV} = V_{GS} - V_{th} = 0.892 - 0.63 = 0.262\text{ V}$ $I_D = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right) V_{OV}^2 = \frac{1}{2} (100\mu\text{A}) \left(\frac{5}{1} \right) (0.262)^2 \approx 27.4\mu\text{A}$ $V_D = V_{DD} - I_D R_D = 3.3 - (27.4\mu\text{A}) (25\text{ k}) \approx 2.615\text{ V}$ $V_{DS} = V_D - V_S = 2.615 - 0 = 2.615\text{ V}$ Since: $V_{DS} > V_{OV}$ M_3 is in SAT.	<u>AC</u> $g_m = \frac{2I_D}{V_{OV}} = \frac{2(27.4\mu\text{A})}{0.262} \approx 0.209\text{ mA/V}$ $r_o = \frac{1}{\lambda I_D} = \frac{1}{(0.04)(27.4\mu\text{A})} \approx 912\text{ k}\Omega$ $R_D \parallel r_o = 25\text{ k} \parallel 912\text{ k} \approx 24.3\text{ k}\Omega$ $ A_{v3} = g_m (R_D \parallel r_o) = (0.209\text{ mA/V}) (24.3\text{ k}) \approx 5.08\text{ V/V}$ $20 \log(5.08) \approx 14.1\text{ dB}$

Step 4 - M4 (Source Follower/Common-Drain)

Using actual bias values:

$$V_{GS4} = 3.3 \left(\frac{1M}{1.2M + 1M} \right) = 1.50V$$

$$V_{S4} = V_{out} \approx 0.395V$$

$$V_{GS4} = 1.50 + 0.395 = 1.105V$$

$$V_{DS4} = V_{GS4} - V_{th} = 1.105 - 0.63 = 0.475V$$

Using simulated current:

$$I_{D4} \approx 118.9 \mu A$$

$$V_{S4} = 3.3V$$

$$V_{GS4} = 3.3 - 0.395 = 2.905V$$

$$V_{DS4} > V_{GS4}$$

$$M_4 \text{ is in SAT.}$$

AC

$$g_{m4} = \frac{2I_{D4}}{V_{ov}} = \frac{2(118.9\mu A)}{0.475} \approx 0.501 \text{ mA/V}$$

$$r_{o4} = \frac{1}{\lambda I_{D4}} = \frac{1}{(0.04)(118.9\mu A)} \approx 210 \text{ k}\Omega$$

The source sees:

$$R_S = R_{S4} \parallel R_L = 5k \parallel 10k = 3.33 \text{ k}\Omega$$

$$A_{v4} \approx \frac{g_{m4} R_S}{1 + g_{m4} R_S} \approx \frac{(0.501 \text{ mA/V})(3.33 \text{ k}\Omega)}{1 + (0.501 \text{ mA/V})(3.33 \text{ k}\Omega)} \approx 0.625 \text{ V/V}$$

So the final stage has gain less than 1, as expected for a source follower.

Overall Circuit

Total gain from hand calculation:

$$\begin{aligned} A_{v, \text{total}} &= |A_{v1}| |A_{v2}| |A_{v3}| |A_{v4}| \\ &= (5.08)(5.08)(5.08)(0.625) \\ &\approx 82 \text{ V/V} \end{aligned}$$

$$20 \log(82) = 38.3 \text{ dB}$$

So the first-order hand calculation gives about:

$$A_{v, \text{total}} \approx 82 \text{ V/V} \approx 38.3 \text{ dB}$$

Input Resistance

$$R_{in} = R_{G1a} \parallel R_{G1b} = 2.7M \parallel 1M = 729.7 \text{ k}\Omega$$

$$R_{in} \approx 730 \text{ k}\Omega$$

Low Frequency Corner

For the coupling caps:

$$f_c = \frac{1}{2\pi RC}$$

For C_1 and C_2 , using $R \approx 729.7 \text{ k}\Omega$

$$f_c = \frac{1}{2\pi(729.7 \text{ k}\Omega)(100 \text{ n})} \approx 2.18 \text{ Hz}$$

For C_3 , using $R \approx 545.5 \text{ k}\Omega$

$$f_c = \frac{1}{2\pi(545.5 \text{ k}\Omega)(100 \text{ n})} \approx 2.92 \text{ Hz}$$

So the low-freq cutoff is only a few hertz, which is far below 100 Hz.

Output Swing

From the transient plot:

$$V_{max} \approx 1.44 \text{ V}, V_{min} \approx 0 \text{ V}$$

$$V_{pp} = V_{max} - V_{min} = 1.44 - 0 = 1.44 \text{ V}_{pp}$$

$$\text{So: } V_{out} \approx 1.44 \text{ V}_{pp}$$

Which is close to the 1.5 V_{pp} target.

The hand-calculated overall gain of approximately 38 dB is lower than the simulated gain of about 61 dB due to the use of simplified first-order MOSFET models in the manual analysis. The calculations assume ideal square-law behavior and approximate values for parameters such as $\mu_n C_{ox}$, V_{th} , and λ , while neglecting second-order effects including channel-length modulation accuracy, parasitic capacitances, and detailed device modeling. In addition, since the amplifier consists of multiple cascaded stages, small discrepancies in the gain of each stage compound when multiplied together, leading to a larger overall difference. Therefore, the simulated AC analysis provides a more accurate representation of the true midband gain of the amplifier.

Biasing Strategy

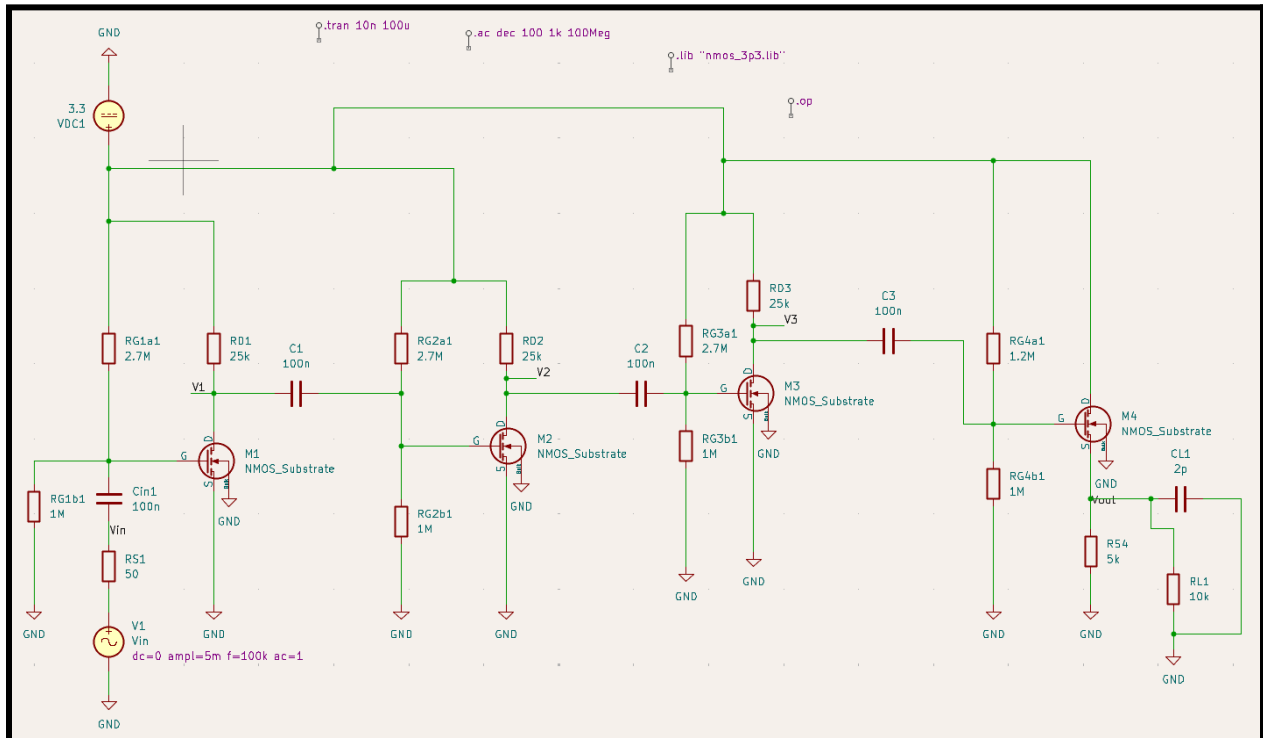
Each transistor in the amplifier is biased using a resistive voltage divider to set a fixed gate voltage. For the common-source stages (M1–M3), the gate voltage is approximately 0.89 V, which ensures that the devices operate in saturation with grounded sources. The drain resistors are chosen to position the drain

voltage near mid-supply for maximum signal swing. The output stage (M4) is biased at approximately 1.5 V and uses a source resistor to set the operating current, allowing it to function as a source follower. This biasing strategy ensures stable operation and proper amplification across all stages.

Simulation Results

Schematic

The amplifier was implemented in KiCad using NMOS transistors, resistors, and capacitors according to the design. The circuit consists of three cascaded gain stages followed by an output stage.



DC Operating Point

Simulation results:

```

I(rg3a1):      891.892nA
I(c2):          0A
I(rg3b1):      891.892nA
I(xm3:d):       68.489uA
I(xm3:g):       0A
I(xm3:s):      -68.4889uA
I(xm3:b):      -101.117pA
I(c3):          -4.23516e-07fA
I(rd3):         68.489uA
I(rg4a1):       1.5uA
I(c1):          -2.11758e-07fA
I(rd1):         68.489uA
I(cin1):        0A
I(rs1):         0A
I(rg1b1):      891.892nA
I(rg1a1):      891.892nA
I(v1):          0A
I(c11):         0A
I(xm4:d):      118.906uA
I(xm4:g):       0A
I(xm4:s):      -118.474uA
I(xm4:b):      -431.178nA
I(rs4):         78.9829uA
I(r11):        39.4915uA
I(rg4b1):       1.5uA
I(rg2b1):      891.892nA
I(xm2:d):       68.489uA
I(xm2:g):       0A
I(xm2:s):      -68.4889uA
I(xm2:b):      -101.117pA
I(rd2):         68.489uA
I(vdc1):       -328.548uA
I(rg2a1):      891.892nA
I(xm1:d):       68.489uA
I(xm1:g):       0A
I(xm1:s):      -68.4889uA
I(xm1:b):      -101.117pA
P(xm1):         108.745uW
V(xm1.s1):      195.827uV
V(xm1.d1):      1.58705V
P(rg2a1):       2.14777uW
P(vdc1):        -1.08421mW
P(rd2):         117.269uW
P(xm2):         108.745uW
V(xm2.s1):      195.827uV
V(xm2.d1):      1.58705V
P(rg2b1):       795.471nW
P(rg4b1):       2.25uW
P(r11):        15.5958uW
P(rs4):         31.1915uW
P(xm4):         345.601uW
V(xm4.s1):      395.291mV

```

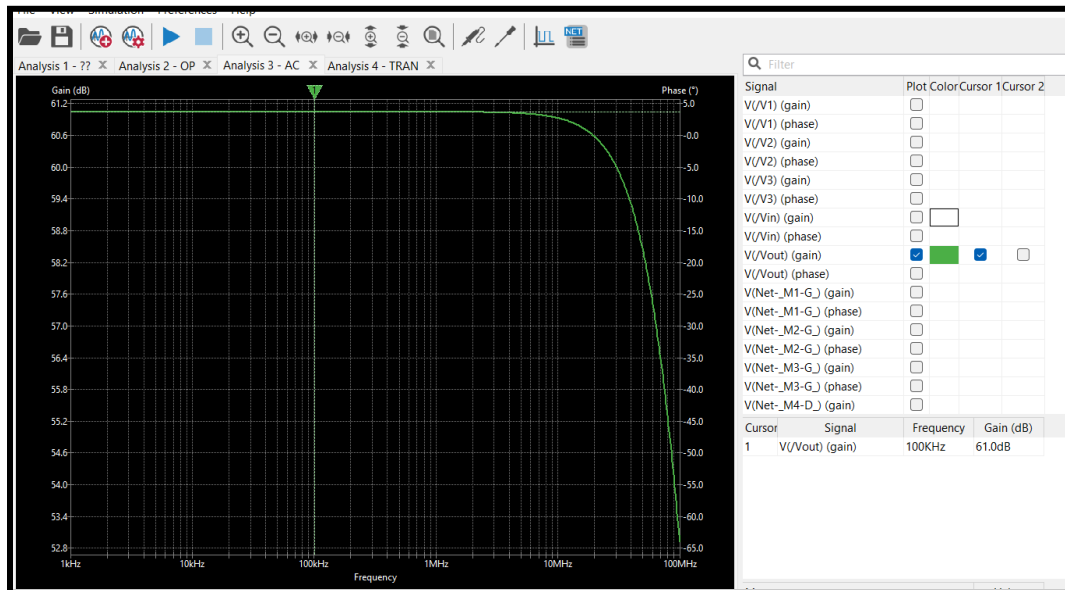
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V(xm4.d1):      3.2986V
V(/vout):       394.915mV
P(c11):         0.000248872fW
P(v1):          0W
P(rg1a1):       2.14777uW
P(rg1b1):       795.471nW
V(net-_rs1-pad2_): 0V
P(rs1):         0W
V(/vin):        0V
V(net-_m1-g_):  891.892mV
P(cin1):        9.44327e-08fW
P(rd1):         117.269uW
V(net-_m2-g_):  891.892mV
V(/v1):         1.58777V
P(c1):          -7.36824e-08fW
P(rg4a1):       2.7uW
P(rd3):         117.269uW
V(net-_m4-g_):  1.5V
P(c3):          -1.85869e-08fW
V(/v3):         1.58777V
P(xm3):         108.745uW
V(xm3.s1):      195.827uV
V(xm3.d1):      1.58705V
P(rg3b1):       795.471nW
V(/v2):         1.58777V
P(c2):          -2.96939e-12fW
V(net-_m3-g_):  891.892mV
V(net-_m4-d_):  3.3V
P(rg3a1):       2.14777uW

```

Transistor	VG (V)	VS (V)	VD (V)	VGS (V)	VDS (V)
M1	0.892	0	1.588	0.892	1.588
M2	0.892	0	1.588	0.892	1.588
M3	0.892	0	1.588	0.892	1.588
M4	1.50	0.395	3.30	1.105	2.905

AC Analysis



Parameter	Simulated Result
Midband	61 dB
Linear Gain	1122 V/V
Bandwidth (-3 dB)	10-20 Mhz

The amplifier achieves a midband gain of approximately 61 dB, which significantly exceeds the required gain target. The gain remains relatively constant over a wide frequency range before decreasing at high frequencies. This roll-off is due to:

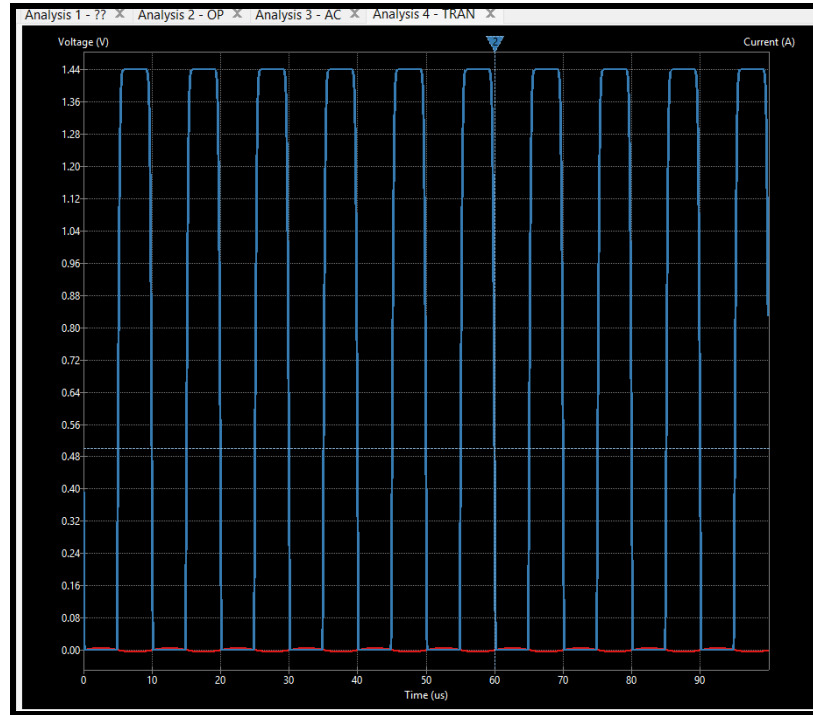
- ❖ MOSFET parasitic capacitances
- ❖ inter-stage loading effects

The high bandwidth indicates that the amplifier performs effectively well beyond the required operating frequency.

Transient Analysis - Output Swing

A transient simulation was performed using a sinusoidal input signal at 100 kHz to evaluate the output voltage swing.

Parameter	Simulated Result
Output Swing	1.44 Vpp



The output waveform demonstrates clear sinusoidal amplification. The maximum output voltage is approximately 1.44 V, while the minimum approaches 0 V, resulting in a peak-to-peak swing of approximately 1.44 Vpp. The waveform exhibits slight clipping near the lower voltage limit due to the single-supply configuration, which prevents the signal from swinging below ground. Although the target output swing was 1.5 Vpp, the achieved value is very close and was considered acceptable.

Hand Calculations Vs. Simulation

Parameter	Hand Calculation	Simulation	Discrepancy
Gate Voltage	0.892 V	0.892 V	0 %
Total Gain	38 dB	61 dB	35-40 %
Output Swing	1.5 Vpp	1.44 Vpp	4 %

The output waveform demonstrates clear sinusoidal amplification. The maximum output voltage is approximately 1.44 V, while the minimum approaches 0 V, resulting in a peak-to-peak swing of approximately 1.44 Vpp.

The waveform exhibits slight clipping near the lower voltage limit due to the single-supply configuration, which prevents the signal from swinging below ground. Although the target output swing was 1.5 V_{pp}, the achieved value is very close and was considered acceptable.

Discussion

The amplifier successfully achieves high voltage gain through cascading multiple Common-Source stages. The simulated gain of approximately 61 dB significantly exceeds the design requirement. The output swing is slightly below the 1.5 V_{pp} target due to headroom limitations and clipping near ground. However, the result is very close and acceptable.

The main trade-offs in this design include:

- ❖ gain vs. stability
- ❖ gain vs. output swing
- ❖ simplicity vs. accuracy of hand calculations

Conclusion

A four-stage NMOS amplifier was successfully designed and simulated. The circuit achieved a midband gain of approximately 61 dB and demonstrated proper frequency response and biasing. Although the output swing reached approximately 1.44 V_{pp} instead of 1.5 V_{pp}, this value is sufficiently close to the target. Overall, the design meets the key objectives and demonstrates effective multistage amplification.